



Ho Chi Minh City University of Technology
Faculty of Computer Science and Engineering

Algorithms Analysis and Design

(C03031)

Computer Science Program

Instructor: Assoc. Prof. Dr. Vo Thi Ngoc Chau

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Course Learning Outcomes

1. Able to analyze the complexity of the algorithms (recursive or iterative) and estimate the efficiency of the algorithms.
2. Improve the ability to design algorithms in different areas *by decomposing a computing problem into advanced algorithm design strategies.*
3. Able to discuss NP-completeness.



References

- [1] Cormen, T. H., Leiserson, C. E, and Rivest, R. L., *Introduction to Algorithms*, The MIT Press, 2009.
- [2] Levitin, A., *Introduction to the Design and Analysis of Algorithms*, 3rd Edition, Pearson, 2012.
- [3] Sedgewick, R., *Algorithms in C++*, Addison-Wesley, 1998.
- [4] Weiss, M.A., *Data Structures and Algorithm Analysis in C*, TheBenjamin/Cummings Publishing, 1993.



Course Outline

- ❑ Chapter 1. Fundamentals
- ❑ Chapter 2. Divide-and-Conquer Strategy
- ❑ Chapter 3. Decrease-and-Conquer Strategy
- ❑ Chapter 4. Transform-and-Conquer Strategy
- ❑ Chapter 5. Dynamic Programming and Greedy Strategies
- ❑ Chapter 6. Backtracking and Branch-and-Bound
- ❑ Chapter 7. NP-completeness
- ❑ Chapter 8. Approximation Algorithms



Chapter 1. Fundamentals

- 1.1. Recursion and recurrence relations
- 1.2. Analysis of algorithms
- 1.3. Analysis of iterative algorithms
- 1.4. Analysis of recursive algorithms
- 1.5. Algorithm design strategies
- 1.6. Brute-force algorithm design



Chapter 2.

Divide-and-Conquer Strategy

- 2.1. Divide-and-conquer strategy
- 2.2. Quicksort
- 2.3. Mergesort
- 2.4. External sorting
- 2.5. Binary search tree



Chapter 3.

Decrease-and-Conquer Strategy

- ❑ 3.1. Decrease- and-conquer strategy
- ❑ 3.2. Insertion sort
- ❑ 3.3. Graph traversal algorithms
- ❑ 3.4. Topological sorting
- ❑ 3.5. Generating all permutations from a set



Chapter 4. Transform-and-Conquer Strategy

- 4.1. Transform-and-conquer strategy
- 4.2. Gaussian Elimination for solving a system of linear equations
- 4.3. Heaps and heapsort
- 4.4. Horner's rule for polynomial evaluation
- 4.5. String matching by Rabin-Karp algorithm



Chapter 5.

Dynamic Programming and Greedy Strategies

□ 5.1. Dynamic Programming

- 5.1.1. Matrix-chain multiplication
- 5.1.2. Longest common subsequence
- 5.1.3. (0-1) Knapsack problem
- 5.1.4. Warshall algorithm to find transitive closure
- 5.1.5. Floyd algorithm for the all-pairs shortest path problem

□ 5.2. Greedy Algorithms

- 5.2.1. Activity-selection problem
- 5.2.2. Fractional knapsack problem
- 5.2.3. Huffman codes
- 5.2.4. Graph coloring



Chapter 6.

Backtracking and Branch-and-Bound

- 6.1. Backtracking algorithms
 - 6.1.1. The Knight's Tour problem
 - 6.1.2. The Eight Queens problem
- 6.2. Branch-and-Bound algorithms
 - 6.2.1. Traveling Salesman problem



Chapter 7.

NP-completeness

- 7.1. Deterministic and Nondeterministic Polynomial-time algorithms
- 7.2. NP-completeness
- 7.3. Cook's theorem
- 7.4. The halting problem
- 7.5. Some NP-complete problems
- 7.6. Some approaches to cope with NP-complete problems



Chapter 8.

Approximation Algorithms

- 8.1. Why approximation algorithms?
- 8.2. Vertex cover problem
- 8.3. Set cover problem
- 8.4. Traveling Salesman problem
- 8.5. Scheduling Independent tasks
- 8.6. Bin Packing problem



Grading Scheme

- Midterm exam: 40%
 - Weekly in-class activities: 10%
 - Group and individual tasks
 - *Special situation*: Weekly e-learning activities
 - Midterm in-class exam: 30%
 - References in one A4 sheet
- Final exam: 60%
 - References in one A4 sheet



Grading

- ❑ Never copy from the others
- ❑ Never let the others copy from you
- ❑ Never be absent from class if not necessary
 - In-class time less than 75% is not allowed!
- ❑ Never be absent from examination



Contact

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 - By appointment



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